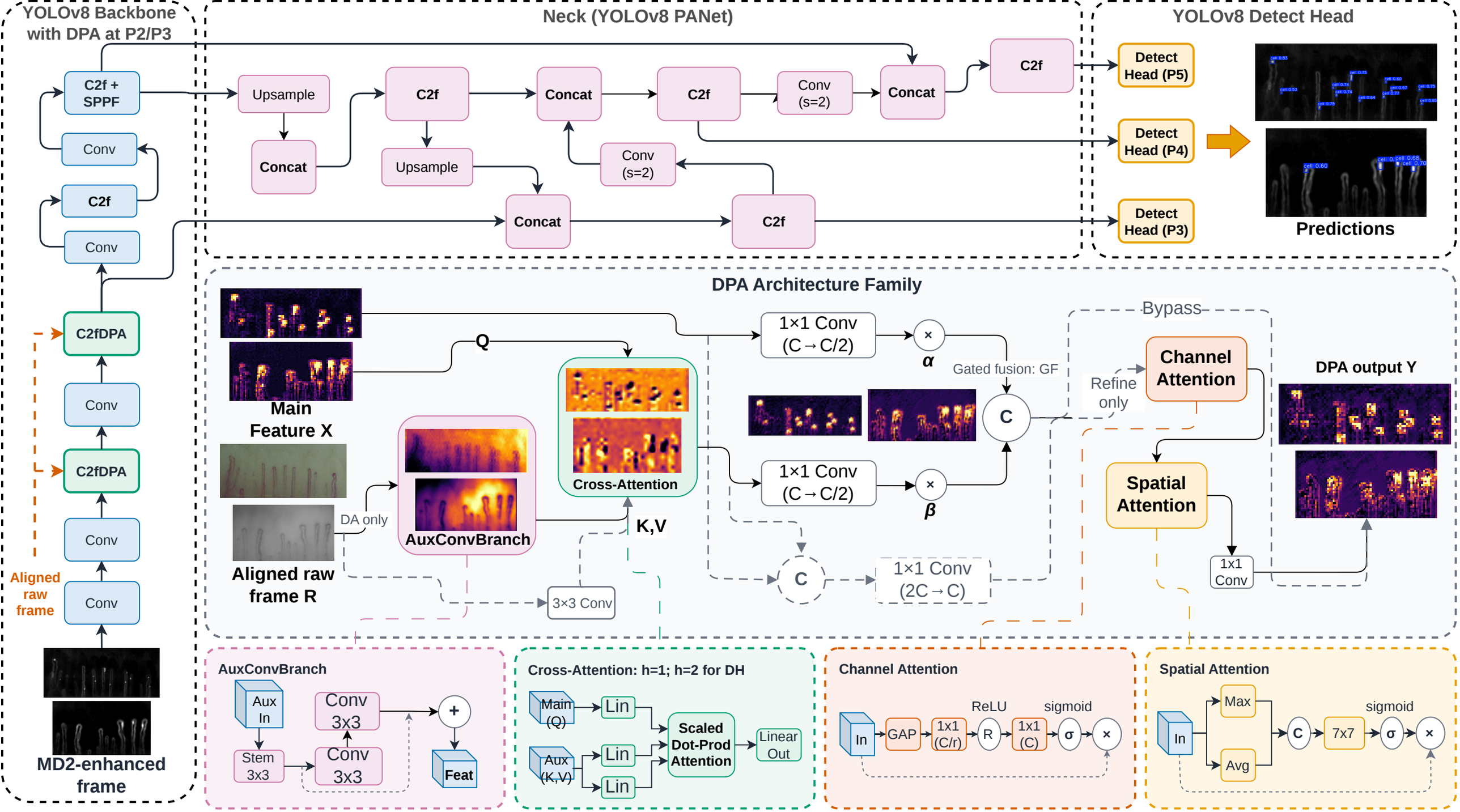

YOLOv8 Backbone with DPA at P2/P3
C2f + SPPF
Conv
C2f
Conv
C2fDPA
Conv
C2fDPA
Conv
Conv
Aligned raw frame
MD2-enhanced frame
Neck (YOLOv8 PANet)
Upsample
Concat
C2f
Upsample
Concat
C2f
Conv (s=2)
Concat
C2f
Concat
Conv (s=2)
C2f
YOLOv8 Detect Head
Detect Head (P5)
Detect Head (P4)
Detect Head (P3)
Predictions
DPA Architecture Family
Main Feature X
Aligned raw frame R
Q
K,V
DA only
AuxConvBranch
Cross-Attention
3×3 Conv
1×1 Conv (C→C/2)
1×1 Conv (C→C/2)
α
β
Gated fusion: GF
C
C
1×1 Conv (2C→C)
Bypass
Refine only
Channel Attention
Spatial Attention
1x1 Conv
DPA output Y
AuxConvBranch
Aux In
Conv 3x3
Stem 3x3
Conv 3x3
Feat
Cross-Attention: h=1; h=2 for DH
Main (Q)
Aux (K,V)
Lin
Lin
Lin
Scaled Dot-Prod Attention
Linear Out
Channel Attention
In
GAP
1x1 (C/r)
ReLU
R
1x1 (C)
sigmoid
σ
Spatial Attention
In
Max
Avg
C
7x7
sigmoid
σ